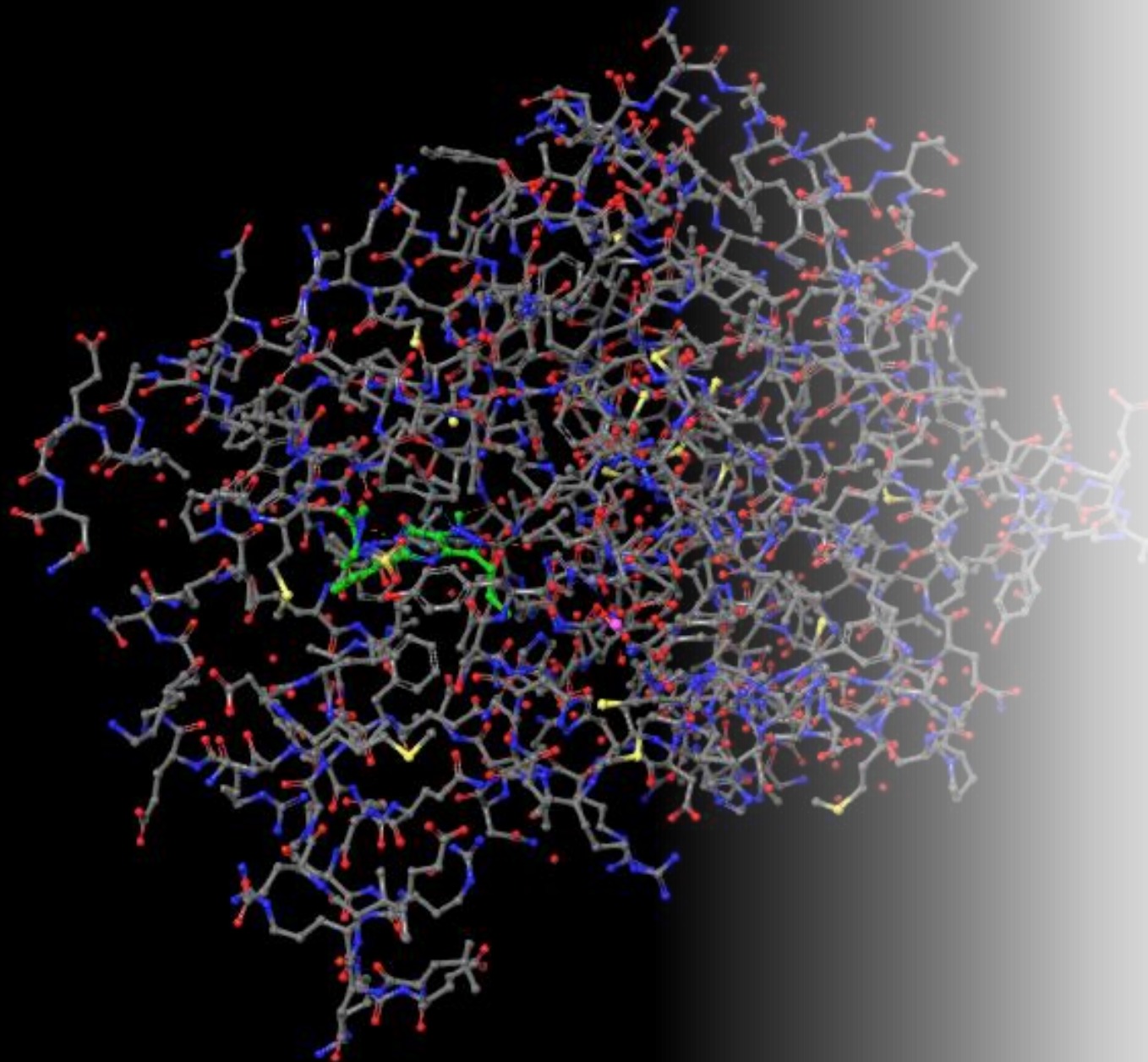
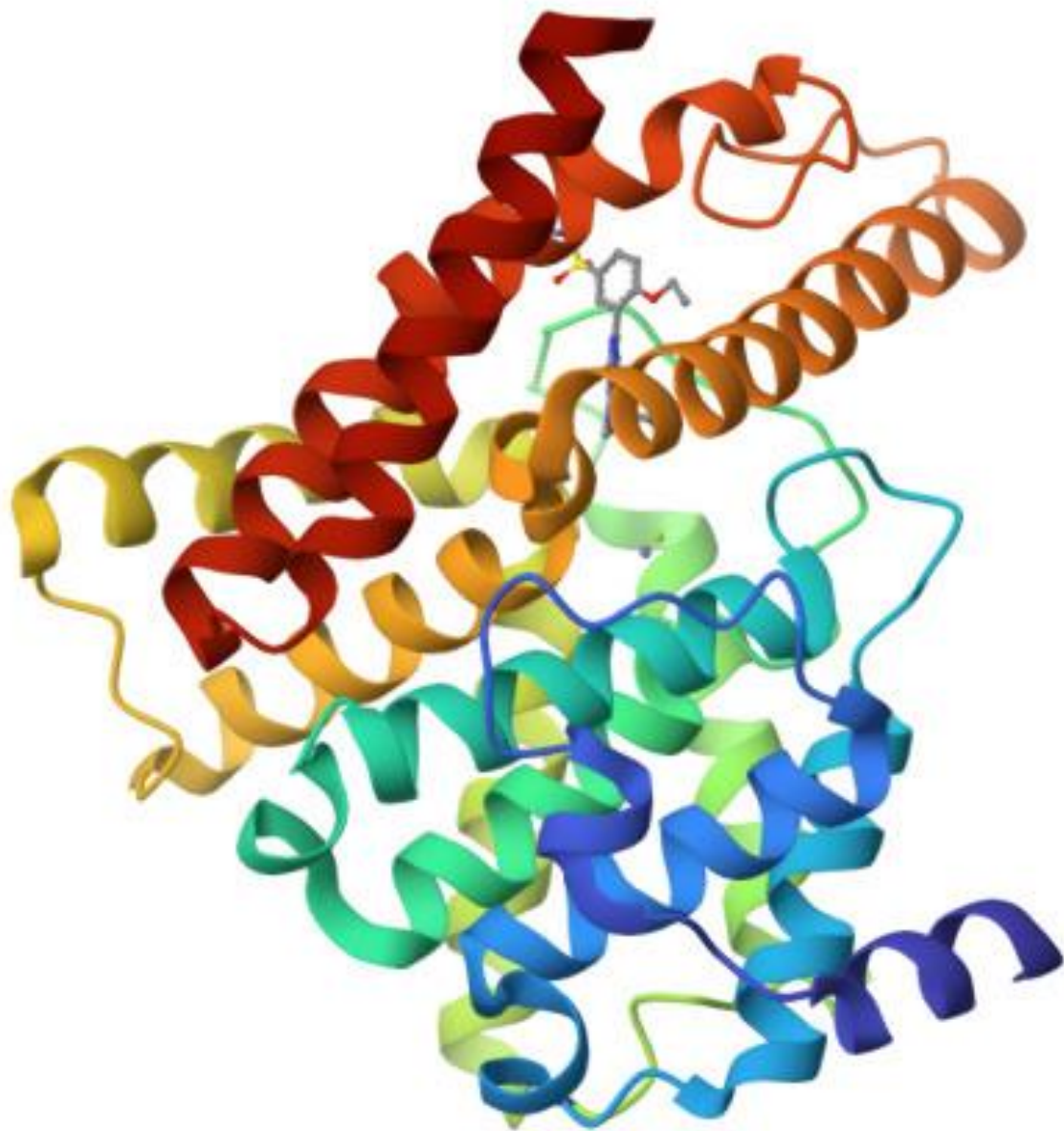




1UDT Group 9
Project:
Sean Cross,
Ewan Sharanda

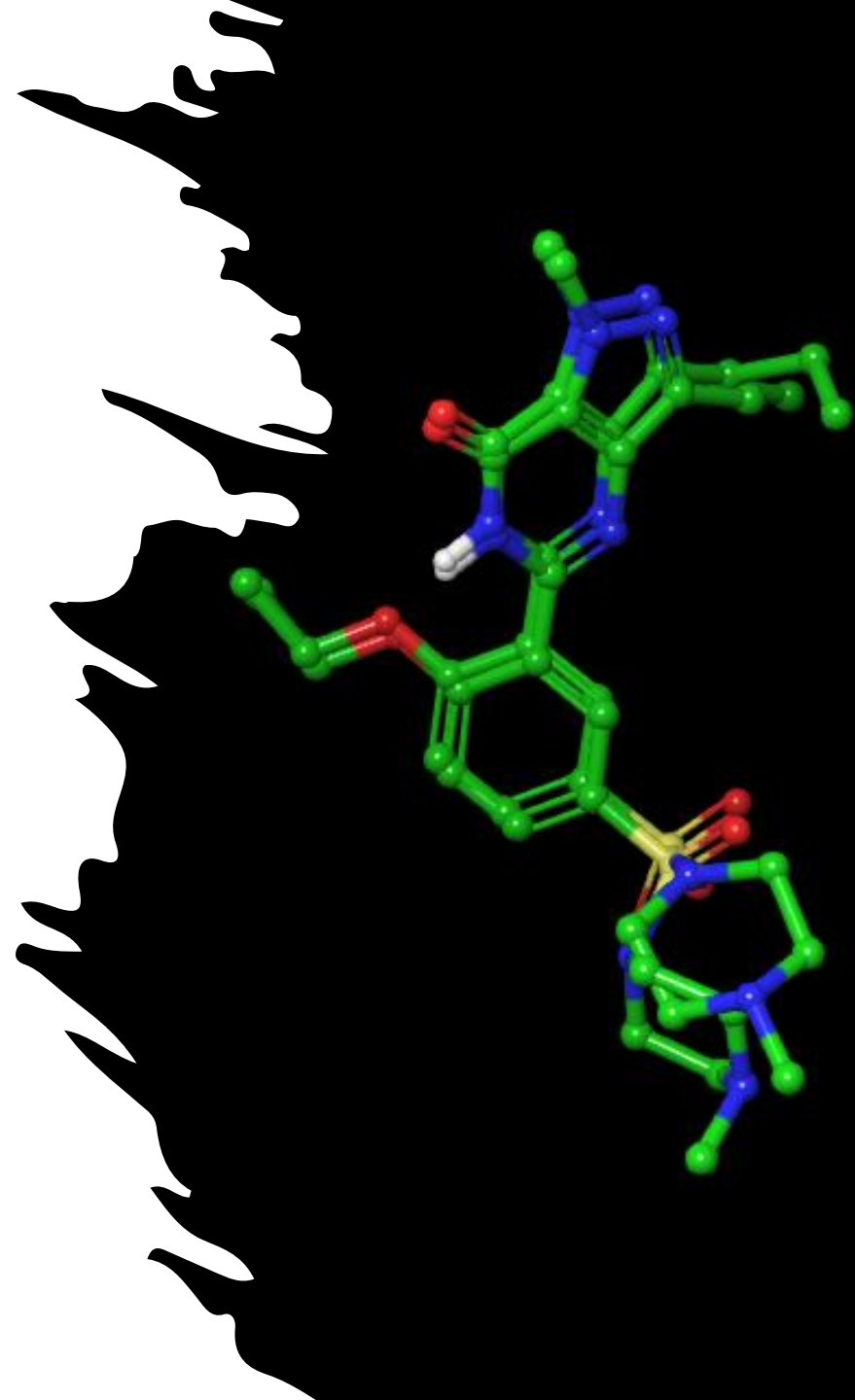


Introduction

- The 1UDT structure is the catalytic domain of the PDE5 enzyme which plays an important role in cellular signalling and muscle regulation. In this project the protein was prepared for docking adjusting water molecules before testing the binding of actives and decoys. Key interactions were studied and a pharmacophore model was developed to identify the essential features required for ligand binding.

Differences in constraints

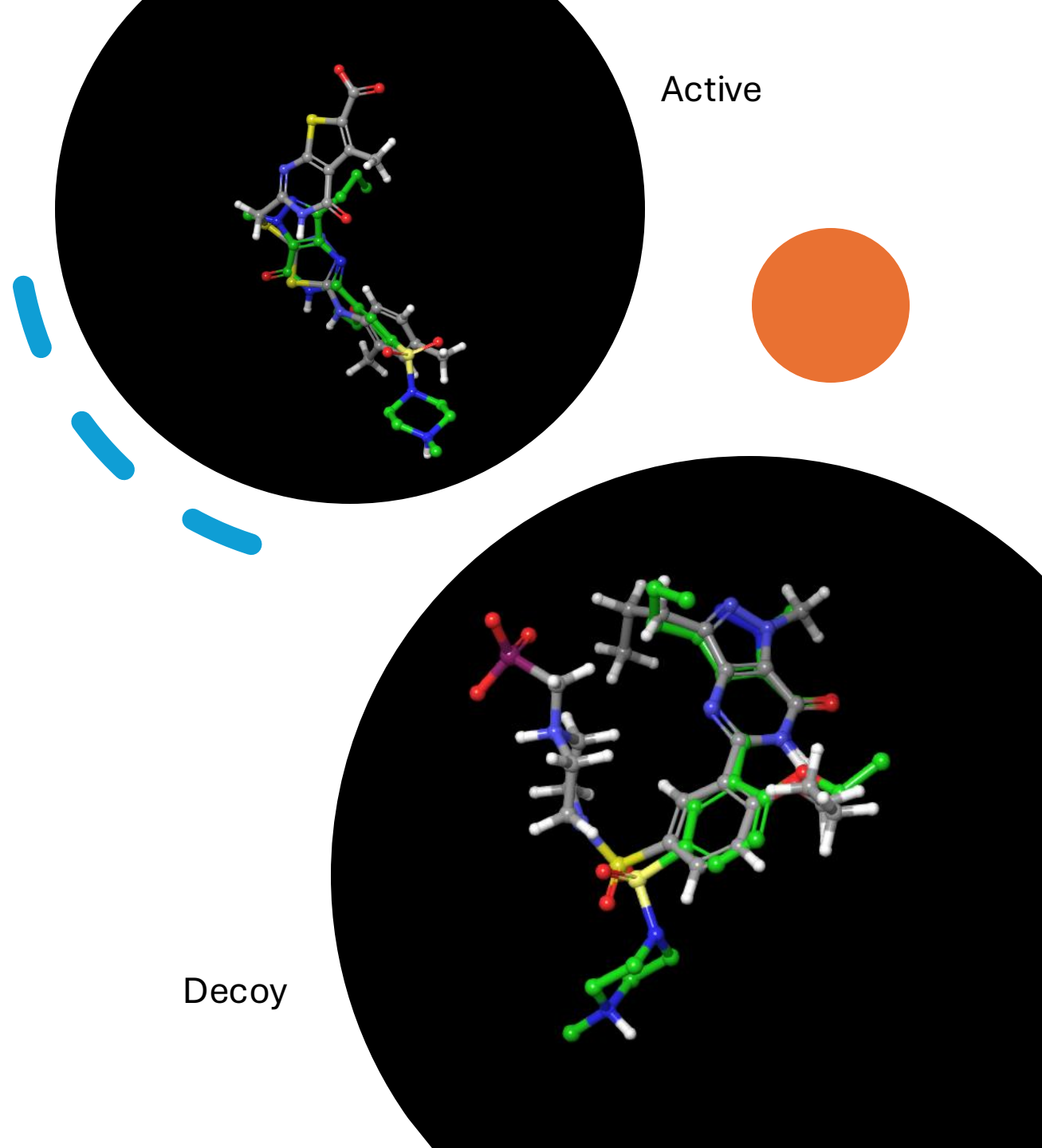
- Sean Cross – Deleted all waters except 1 (H2O:28)
- Ewan Sharanda – Deleted all waters except 2 (H2O: 75, H2O:28)
- Constraints: 817OE1 & 817HE22
- Best docking score ((Sean) = -10.745 (817OE1)
- Best docking score (Ewan) = -10.234 (no constraints)



Docking scores actives & Decoys

Highest
docking active
= -11.612

Highest
docking decoy
= -14.382



Analyzing Docking Scores

- Best decoy scored **better** (−14.382) than best active (−11.612) → poor discrimination
 - Indicates **rigid docking model is not accurately ranking activity**
 - Possible reasons:
 - Binding site may be **too open**, favoring shape-fitting decoys
 - Decoys may accidentally form **strong H-bonds**
 - True actives may require **induced fit**, which Glide cannot capture
 - **Protein flexibility** not represented
 - Missing or incorrect **protonation states** or **key water molecules**
 - Conclusion: Glide scores alone are unreliable → supports need for **pharmacophore or ML-based screening** instead.
-

Ligand interactions

- Analyzed acceptor Tyr(664)
- Analyzed aromatic PHE(820)

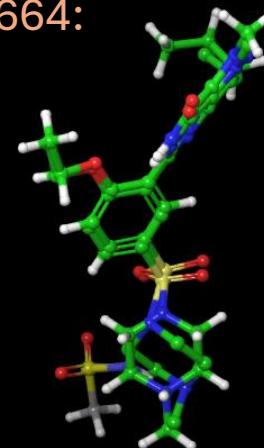
Analysis:

Acceptor docking score = -9.738

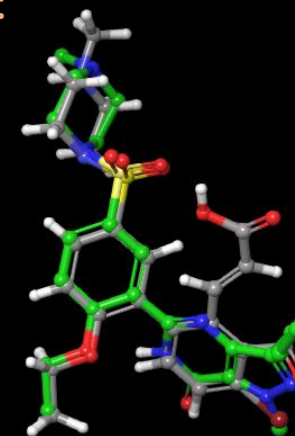
Aromatic docking score = -7.885

- Tyr664 H-bonding contributes more to binding energy than aromatic stacking
- Phe 820 interaction is weaker as pi-pi stacking depends heavily on ligand orientation.
- These findings highlight Tyr664 as a key interaction feature

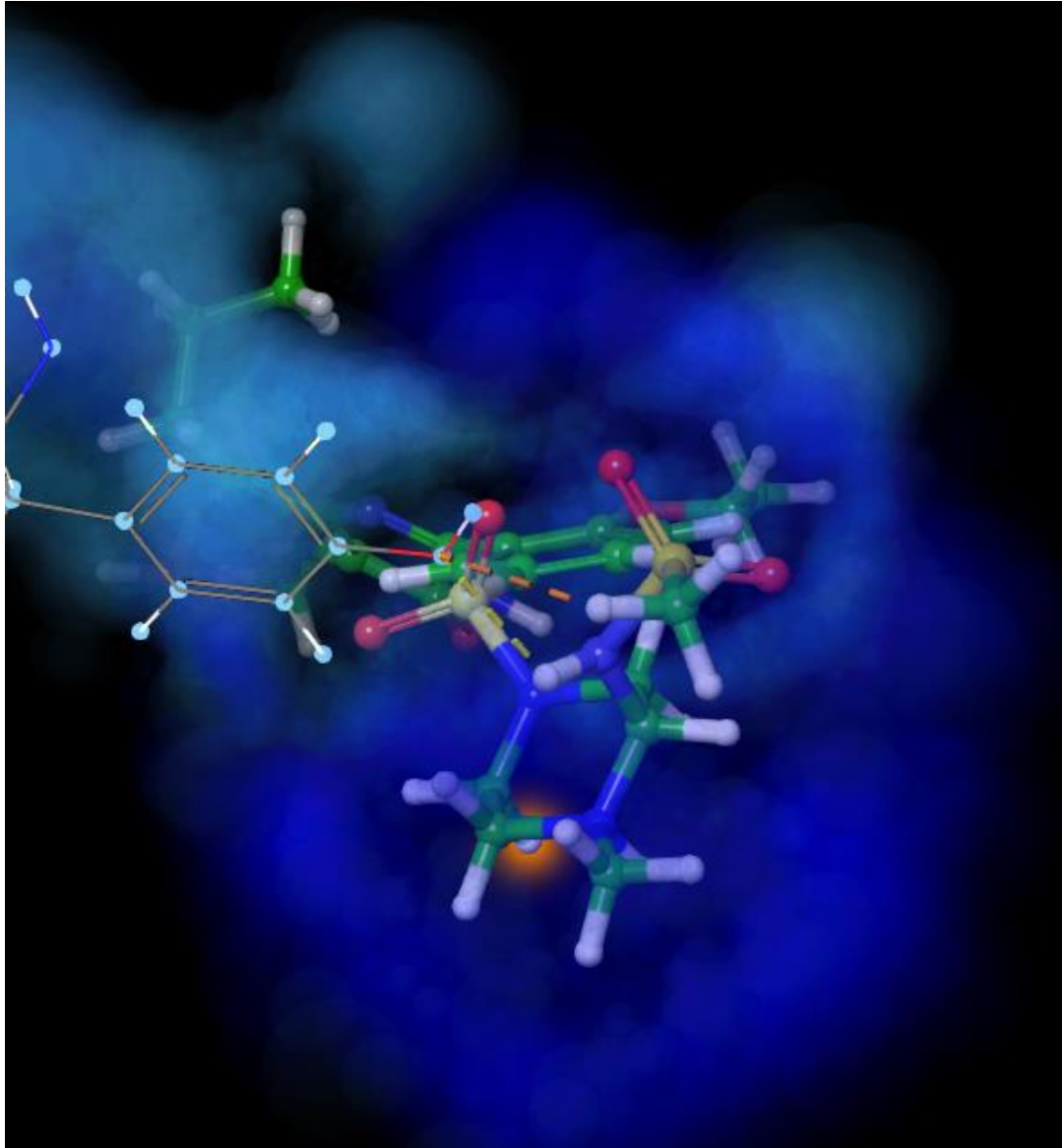
Tyr 664:



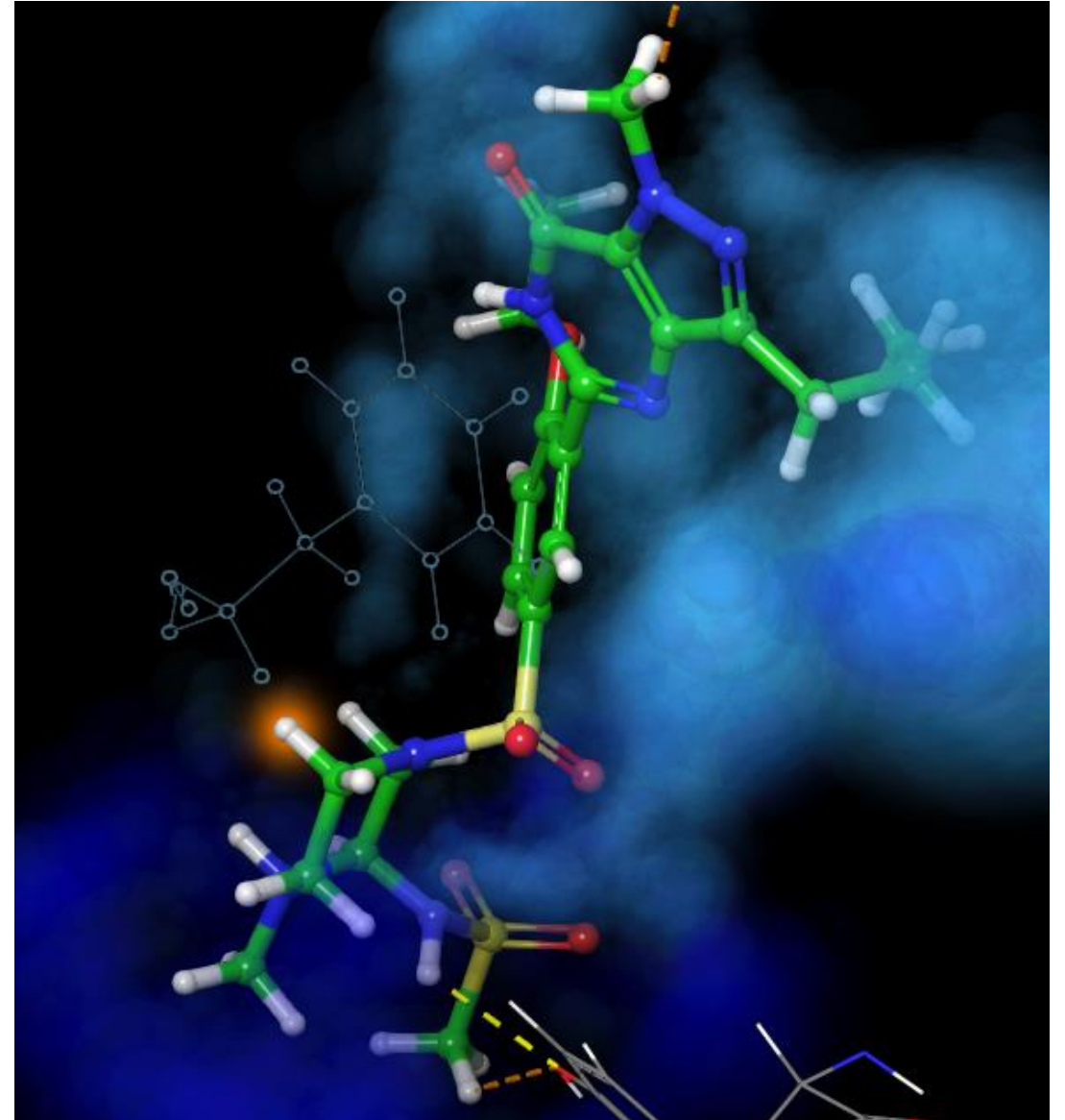
Phe 820:

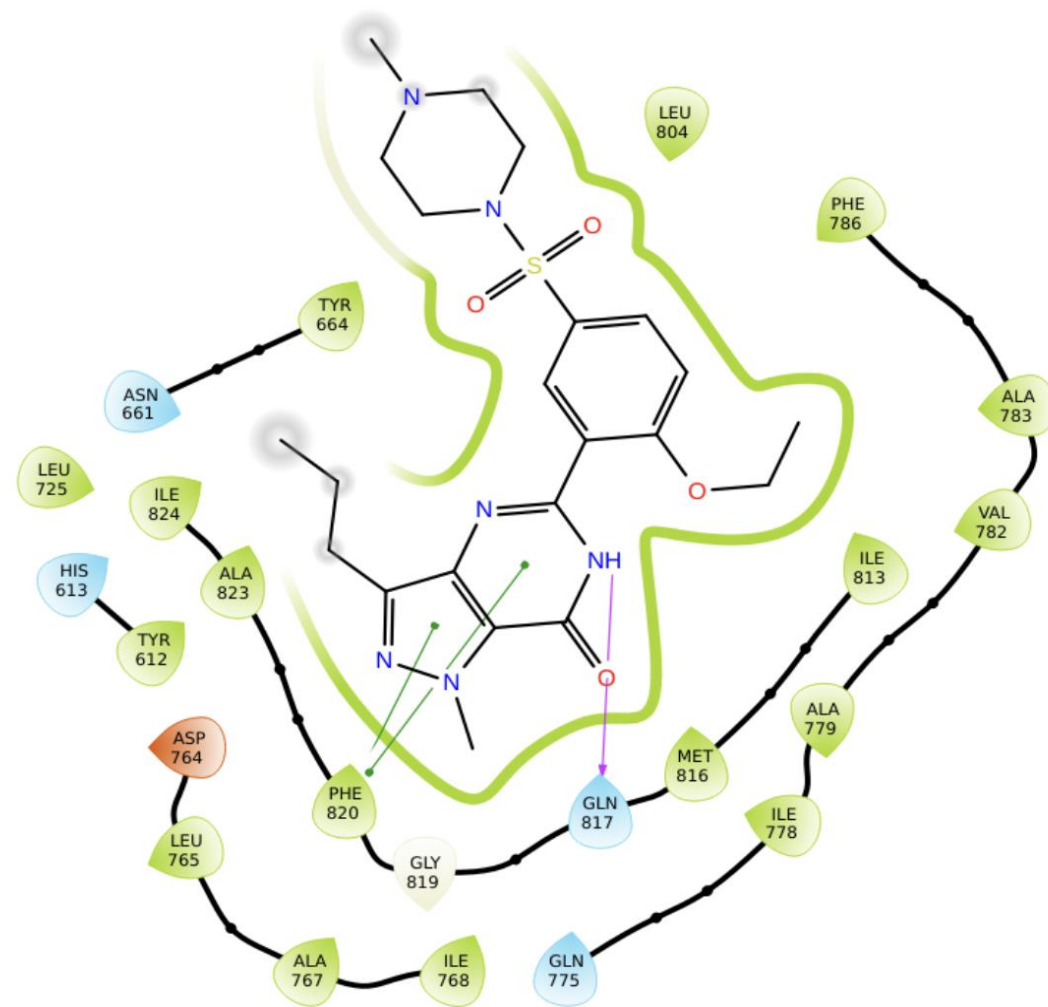


Tyr 664



Phe 820





MPO SCORES

Workspace Navigator

ENTRY LIST

Row	In	Title	mpo score
5752	☐	1UDT - with-deletions_ligand_47	0.842
5753	☐	1UDT - with-deletions_ligand_17	0.831
5754	☐	1UDT - with-deletions_ligand_10	0.830
5755	☐	1UDT - with-deletions_ligand_15	0.762
5756	☐	1UDT - with-deletions_ligand_32	0.740
5757	☐	1UDT - with-deletions_ligand_53	0.734
5758	☐	1UDT - with-deletions_ligand_39	0.730
5759	☐	1UDT - with-deletions_ligand_20	0.730
5760	☐	1UDT - with-deletions_ligand_62	0.725
5761	☐	1UDT - with-deletions_ligand_11	0.720
5762	☐	1UDT - with-deletions_ligand_8	0.717
5763	☐	1UDT - with-deletions_ligand_5	0.714
5764	☐	1UDT - with-deletions_ligand_30	0.713
5765	☐	1UDT - with-deletions_ligand_37	0.712
5766	☐	1UDT - with-deletions_ligand_21	0.712
5767	☐	1UDT - with-deletions_ligand_58	0.710

STRUCTURE HIERARCHY

- ☒ Current Selection
- ☒ All
 - ☒ A: VIA 1000
 - ☒ Ligands
 - ☒ 1UDT - with-deletions
 - ☐ Protein
 - ☒ Solvents
 - ☒ Other

Title: A: VIA 1000
PDB ID: 1UDT

Cavity
Solvent Exposed
Ugly Clash
Bad Clash

Ligand Designer

A: VIA 1000

DISPLAY Workflows

- ☐ Ligand-Receptor Interactions
- ☐ Displaceable Waters
- ☒ Growth Space
- ☐ Ligand Surface
- ☐ Replaceable Waters
- ☐ Pathfinder Bonds
- ☐ Receptor Surface

[View similar purchasable compounds...](#)

MULTI-PARAMETER OPTIMIZATION

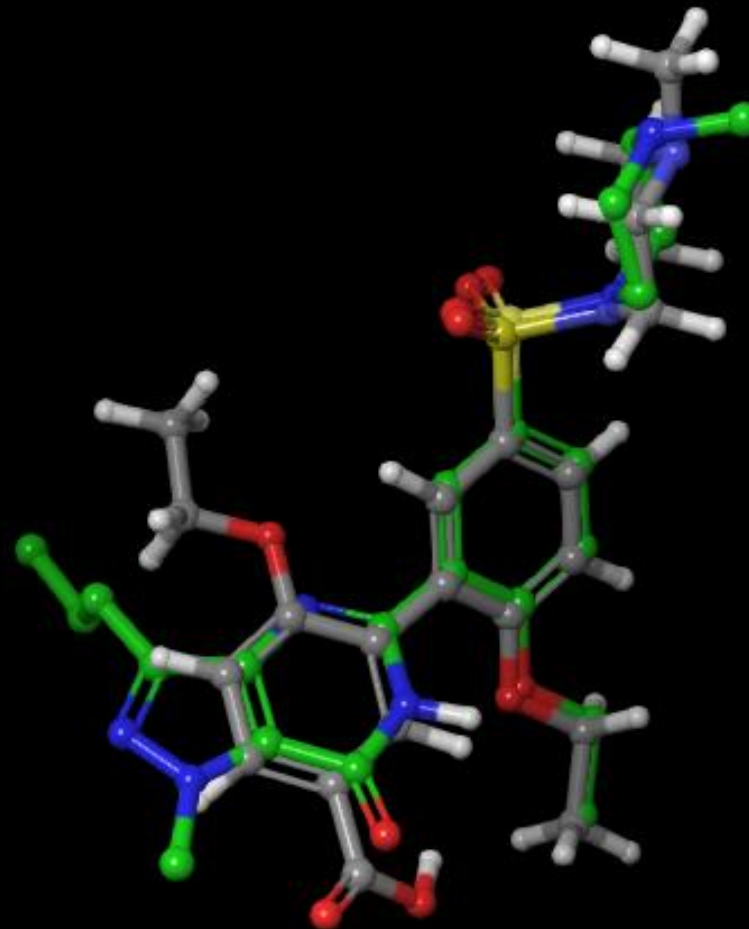
AlogP 1.6 (0.0-5.0) MW 474.6 (0.0-500.0) HBA 8 (0-10) HBD 1 (0-5) PSA 113.4 (0.0-140.0)

POST-PROCESSING

Sort by MPO All Sort

Purchasable product

- Out of 10 similar purchasable products, 7 were successful when attempting to dock with the most successful one having a docking score of -6.836 .



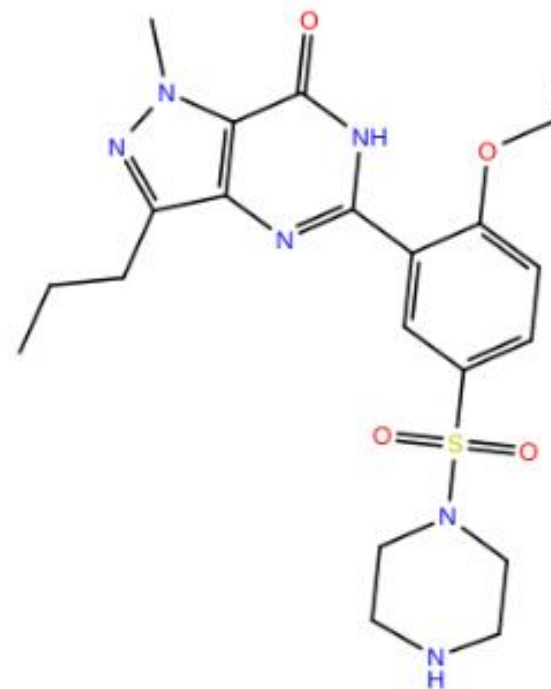
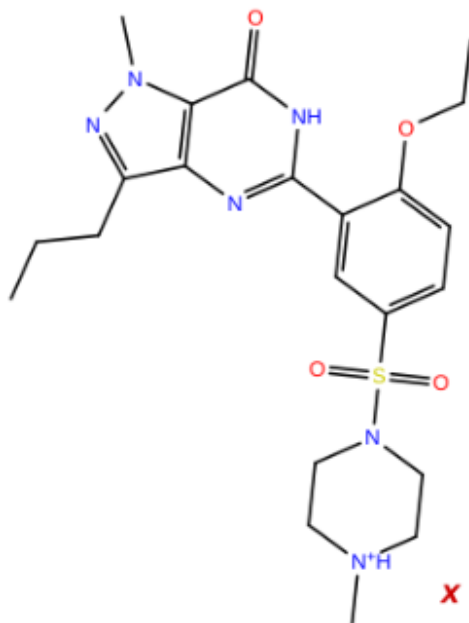


Purchasable Product Analysis

- -6.836 is significantly weaker than -10.745. this suggests that the reference ligand forms key interactions within the active sight that the analogues are unable to replicate. This indicates that only ligands with specific features can bind effectively. These results indicate that the original ligand is well-optimised for the binding pocket, while small structural changes tend to weaken binding rather than improve it.

2D/3D Structure Change

Changed N+H to NH, docking score went from -10.745 to -7.632



2D → 3D Structure Change

Converting N+H with to neutral NH reduced the docking score from **-10.745 to -7.632**

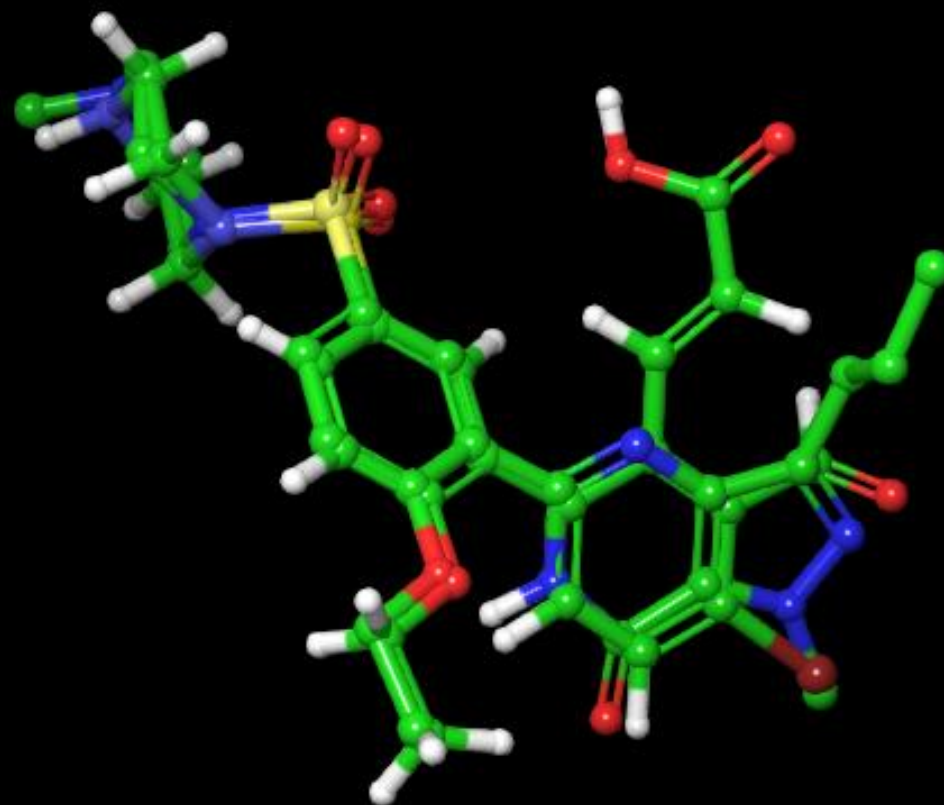
The **positive charge** in the original ligand was important for binding

Charged amine likely formed **stronger electrostatic and H-bond interactions**

Removing the charge **weakened complementarity** in the binding pocket

Shows docking is **highly sensitive to correct protonation states**

Highlights the need to use **biologically relevant ionization** during ligand prep



Pharmacophore Hypothesis

The pharmacophore was prepared using the prepared protein – complex with the deletions of the waters except for 817OE1.



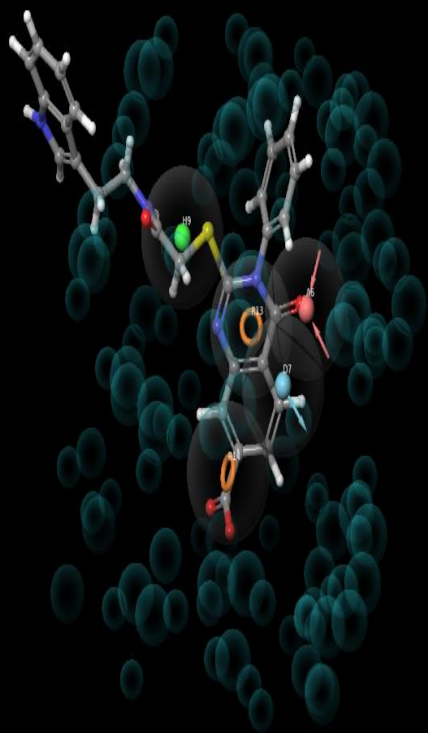
The pharmacophore was developed using the Phe 820 aromatic group interacting with the D29 Donating group.



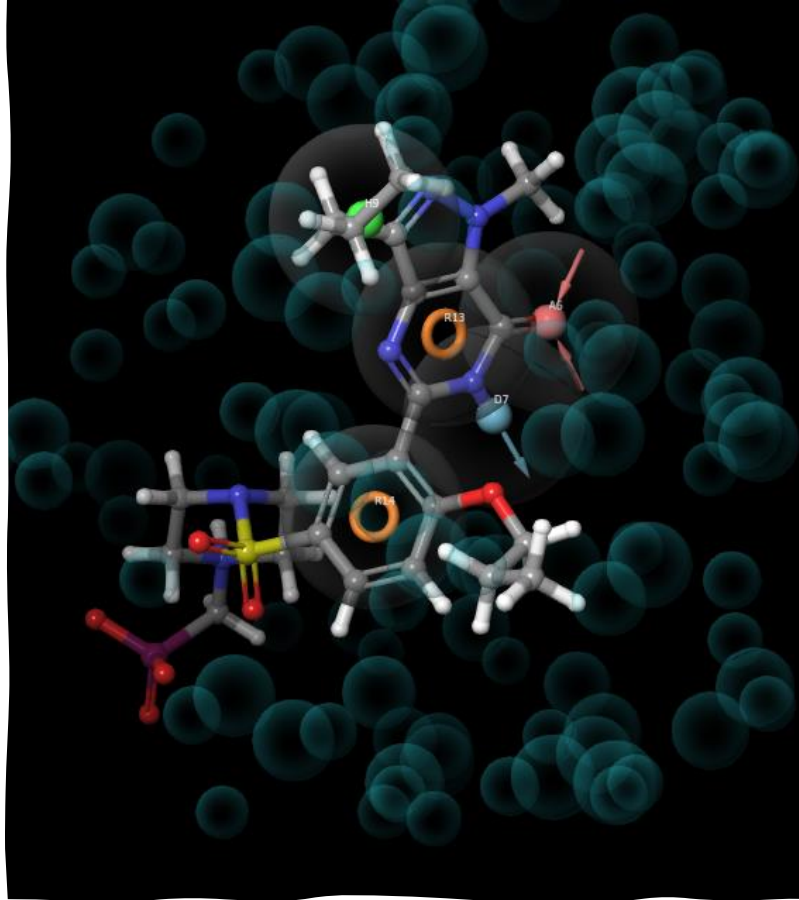
Pharmacophore was then docked to 20 actives and 80 decoys. Their Phase Screen Scores were then recorded.



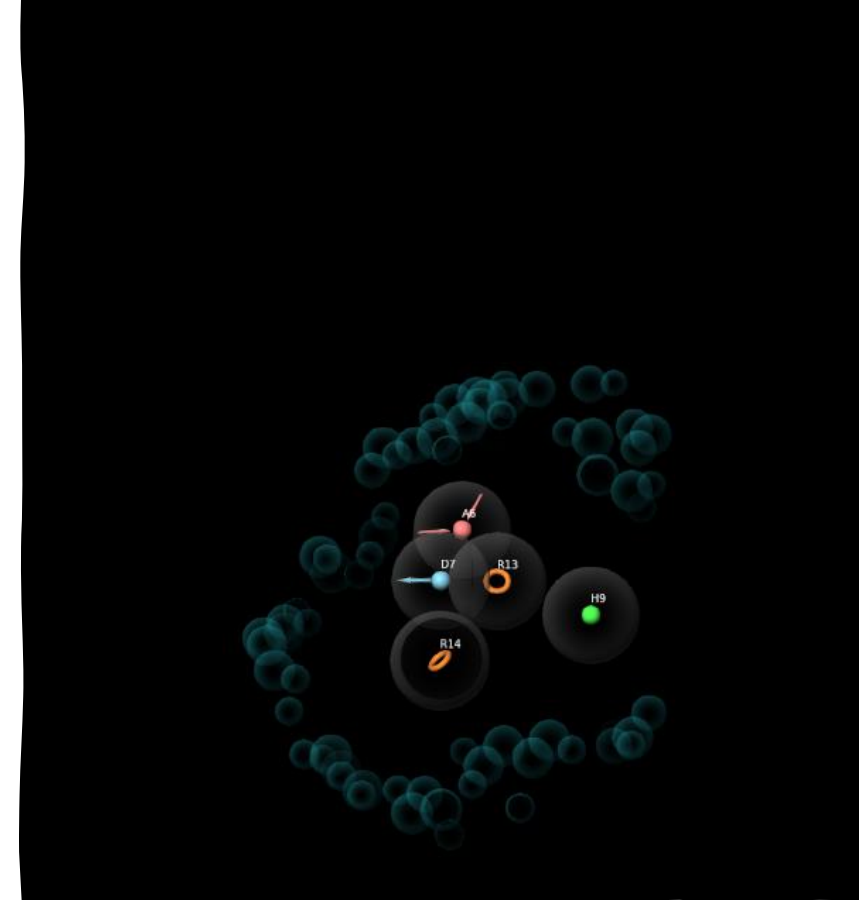
Feature set: R14, R13, D7, A6 and H9



Decoy



Active



Features

Phase Screen Scores

Highest Active: 2.602
Highest Decoy: 1.600
15 actives > highest decoy score

Overall Results

Analysis Category	Key Results
Best Glide Score (Active)	−11.612
Best Glide Score (Decoy)	−14.382 (decoys scored better → poor separation)
Reference Ligand Score	−10.745
Best Purchasable Analogue	−6.836
2D → 3D Protonation Change	Score dropped from −10.745 → −7.632
Key Interactions	Tyr664 (H-bond, −9.738), Phe820 (aromatic, −7.885)
Pharmacophore Features	R14, R13, D7, A6, H9
Highest Phase Score (Active)	2.602
Highest Phase Score (Decoy)	1.600
Actives Scoring Higher Than All Decoys	15 out of 20
Highest MPO Score	0.842